CANARY

Can a computer reading the words in a 10-K rank known fraud filings as anomalous?

A pre-registered single-signal novelty study on SEC 10-K MD&A text.

Primary contribution: a pre-registered null result plus a baseline that beat the autoencoder.

Arsenii Chan · CSC 44800 Artificial Intelligence · Spring 2026 · CCNY

THE DRIVING QUESTION

> *Does an unsupervised autoencoder trained on industry-and-year-matched clean 10-K MD&A text assign elevated reconstruction error to known historical fraud filings, under strictly out-of-sample evaluation?*

The one-line answer this report defends:

At N = 6 frauds, with strict pre-registration: no useful autoencoder signal across four of five evaluable cohorts; Enron is not evaluable under the frozen rule; the one cohort that ranks high (Lehman) is n = 1, and a 1990s TF-IDF baseline ranks it higher.

The contract is the contribution.

git-tagged validation-spec-frozen, committed before any validation script ran.

Out-of-sample, always

Leave-one-cohort-out (LOCO): for each held-out fraud, the autoencoder is trained on clean filings from other cohorts only. The fraud and its peers are never in the training set.

Time-controlled training

Within the LOCO training set, only filings dated on or before that fraud's filing date are eligible. The autoencoder's training corpus cannot include post-fraud disclosure language. (MiniLM pretraining is a separate caveat.)

Single pass, no tuning

Architecture, hyperparameters, sentence cap, seeds: all committed and tagged before validation ran. scripts/06_validate.py runs once. No model tuning after validation.

The contract makes the null result interpretable, not conclusive.

```
git tag: validation-spec-frozen
at 98d2585  ·  2026-05-01 13:42:42 EDT
```

Only Lehman ranks above random expectation; the aggregate still loses to random.

Cohort	Rank	Note
Lehman (LEH, FY 2007) ⚠	**3 / 13**	*n = 1, baseline beats AE*
HealthSouth (HRC)	7 / 12	at or below random
Valeant (VRX)	8 / 13	at or below random
Tyco (TYC)	9 / 13	at or below random
WorldCom (WCOM)	12 / 13	at or below random
Enron (ENE, FY 2000)	—	*not evaluable: filed earliest, no eligible training data*

Aggregate hit-rates

hit@1: **0.00** vs random 0.08

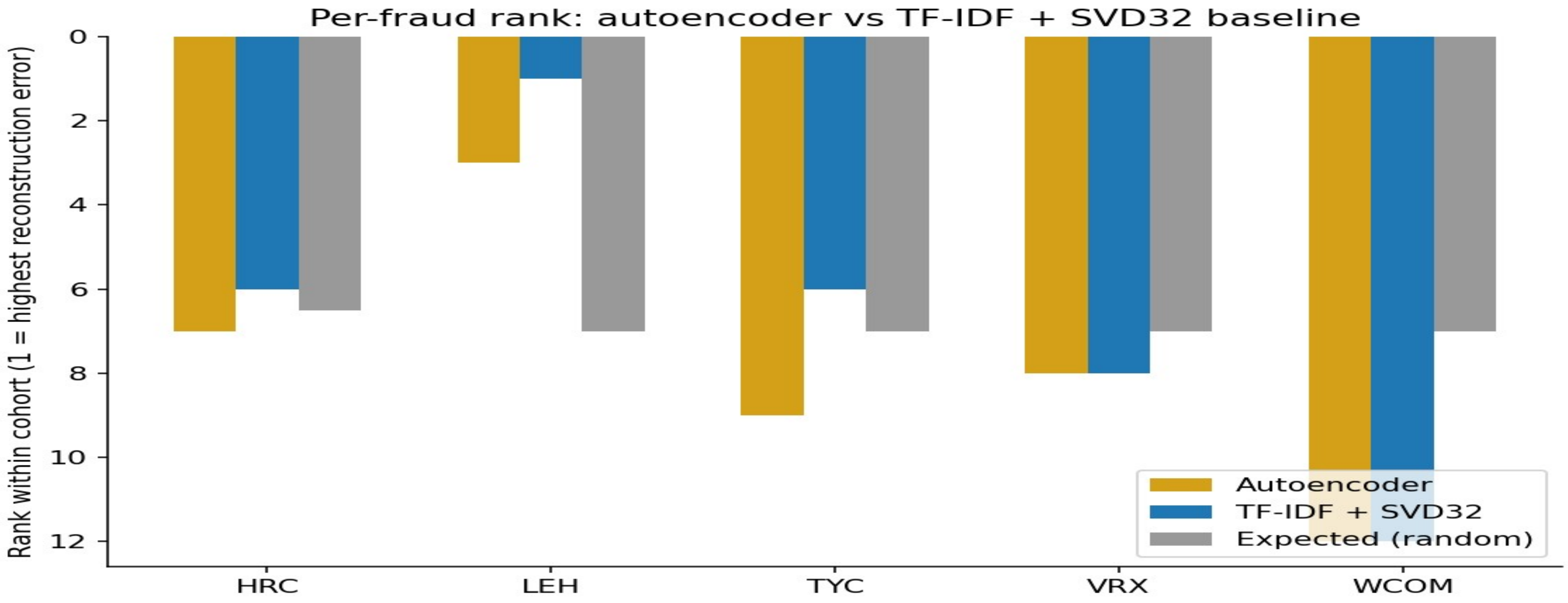
hit@3: **0.20** vs random 0.23

hit@5: **0.20** vs random 0.39

⚠ Lehman is the only rank better than random expectation, and a 1990s baseline ranks it higher (slide 5). The aggregate is entirely Lehman.

A 1990s baseline matched or beat the autoencoder on every cohort.

Post-hoc: tests whether the autoencoder earned its complexity, does not rescue the primary result.



Lehman: AE rank 3/13, TF-IDF rank 1/13. Aggregate hit@5 is 0.40 (TF-IDF) vs 0.20 (autoencoder). Mann-Whitney p is much smaller for TF-IDF on Lehman; exact values are in the report. The pre-registered model does not beat the trivial baseline.

QUESTIONS I EXPECT

Q **Did you check a trivial baseline?**

Yes. TF-IDF + truncated-SVD, same bottleneck dimension, same training corpora. It matches or beats the autoencoder on every cohort. The honest claim is that the autoencoder adds nothing here. Slide 5.

Q **The baseline is post-hoc. Isn't that HARKing (hypothesizing after results)?**

The pre-registered hypothesis was about the autoencoder, and the autoencoder result is reported unchanged. The TF-IDF baseline was added after seeing the null and is reported as post-hoc. The primary null does not depend on it.

Q **Could MiniLM have memorized 'Lehman' / 'Repo 105' / etc.?**

Tested post-hoc. Masked 93 such mentions with [ENTITY] and re-scored. Lehman's rank held at 3/13 (delta = 0). Exact-token name leakage did not explain the rank; topical leakage remains open.

CONCLUSION

What survived: pre-registration.
What failed: the autoencoder beyond the TF-IDF baseline.

A result that loses to a baseline is not evidence for the neural model. The contract held; the model did not.

Next experiment

Same autoencoder, within-firm temporal delta. Does Lehman's MD&A look anomalous against Lehman's own prior-year MD&A?

Pipeline demo (not a fraud scanner): **canary-psi.vercel.app/scan** · Code: **github.com/ArseniiChan/canary** · Tag: **validation-spec-frozen**

Thanks.